SAL COLLEGE OF ENGINEERING

SEMESTER-IV (FOR ALL BRANCHES)

SUBJECT NAME: Environmental Science, Sustainability and Renewable Energy

Code: BE04000101

Mid-Sem Exam Syllabus

Sr. No	Content
1	**INTRODUCTION TO ENVIRONMENT:** ➢ Definition, principles and scope of Environmental Science. Impacts of technology on Environment, Environmental Degradation, Importance of environmental awareness and its conservation across different engineering disciplines
2	**ENVIRONMENTAL POLLUTION** ➢ Water Pollution: Introduction – Water Quality Standards, Sources of Water Pollution, Classification of water pollutants, Effects of water pollutants ➢ Air Pollution: Composition of air, Structure of atmosphere, Ambient Air Quality Standards, Classification of air pollutants, Sources of common air pollutants like PM, SO_2, NO_X, Auto exhaust, Effects of common air pollutants ➢ Noise Pollution: Introduction, Sound and Noise, Noise measurements, Causes and Effects ➢ Land Pollution: Sources, causes, effects, control & Prevention ➢ Solid Waste: Generation and management ➢ Bio-medical Waste: Generation and management ➢ E-waste: Generation and management ➢ Acid Rain, Depletion of Ozone layer and its mitigation.
3	**SUSTAINABILITY:** ➢ Definition, scope of sustainability, ➢ Sustainable development & Circular economy, Sustainable Development goals, ➢ Climate change: Global Warming and Green House Effect: concept and mechanism ➢ Climate Change Effects and Mitigation Strategies ➢ Green Building: Concept, Objectives, Core Principles & Benefits. ➢ Concept of 4R's: Principles, Practical Application of 4R's.